§7.2 Separation of Variables

10/9

Polar coord: $\Delta u = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2} = 0$

For 2nd-order, constant coeff, try Separation of Variables

- Assume a sol'n of the form $u(r,\theta) = R(r)\Theta(\theta)$
- subst in + separate

$$\frac{r}{R}\left[R' + rR''\right] = -\frac{1}{\Theta}\Theta'' = \mu$$

a const.

have to remember that after solving for $R(r)$, $\Theta(\theta)$ when we put back together $u(r,\theta) = \underbrace{R(r)\Theta(\theta)}_{\mu \text{ must be the same.}}$

- Get 2 ODE's. Solve

For R: $r^2R'' + rR' - \mu R = 0$

$$R_\mu(r) = \begin{cases} 1, \ \ln r & \text{if } \mu = 0 \\ r^{\sqrt{\mu}}, \ r^{-\sqrt{\mu}} & \text{if } \mu > 0 \end{cases}$$

For $\Theta(\theta)$: $-\frac{1}{\Theta}\Theta'' = \mu$

$$\Theta'' + \mu\Theta = 0$$

$\mu > 0$

$$\Theta'' + \mu\Theta = 0$$

spring mass system
damping coeff $= 0$

$$\Theta = e^{\lambda\theta}$$

Guess + check

$$\Theta(\theta) = c_1 \cos(\sqrt{\mu}\,\theta) + c_2 \sin(\sqrt{\mu}\,\theta)$$

Try $\Theta = \cos(\mu\theta)$

$\Theta' = -\mu \sin(\mu\theta)$

$\Theta'' = -\mu^2 \cos(\mu\theta)$

$\Theta = \cos(\sqrt{\mu}\,\theta)$
works!

$\mu = 0$: $\Theta'' = 0$ $\Theta' = c_1$ $\Theta = c_1\theta + c_2$

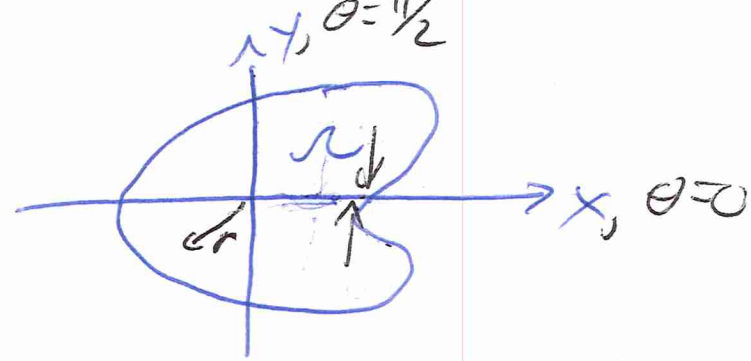
$\mu < 0$? Not considered, because $R(r)$ is not real for $\mu < 0$

$$\Theta_\mu(\theta) = \begin{cases} 1, \ \theta & \mu = 0 \\ \cos(\sqrt{\mu}\,\theta), \ \sin(\sqrt{\mu}\,\theta) & \mu > 0 \end{cases}$$

Now we put together $u(r,\theta) = R_\mu(r)\,\Theta_\mu(\theta)$

must match

But one more thing. To be harmonic $u \in C^2(\Omega)$



suppose Ω contains the origin

For $u \in C^2(\Omega)$ we need

$\Theta_\mu(\theta) = \Theta_\mu(\theta + 2\pi n)$

i.e. sol'n must be periodic!

check $u = \theta$? $0 \neq 2\pi$

$u = \cos(\frac{\theta}{2})$ $1 \neq -1$

This only occurs if $\sqrt{\mu} = n$ (an integer)

So now we have

$$R_n(r) = \begin{cases} 1, \ \ln r & n=0 \\ r^n, \ r^{-n} & n>0 \end{cases}$$

$$\Theta(\theta) = \begin{cases} 1, \ \theta & n=0 \\ \cos(n\theta), \ \sin(n\theta) & n>0 \end{cases}$$

not periodic

$$\boxed{u_n(r,\theta) = \begin{cases} 1, \ r^n\cos(n\theta), \ r^n\sin(n\theta) \\ \ln r, \ r^{-n}\cos(n\theta), \ r^{-n}\sin(n\theta) \end{cases}}$$

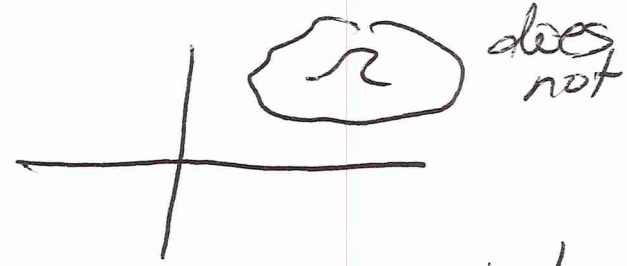
harmonic in entire plane ($\mathbb{R}^2$)

harmonic in $\mathbb{R}^2 \setminus \{(0,0)\}$

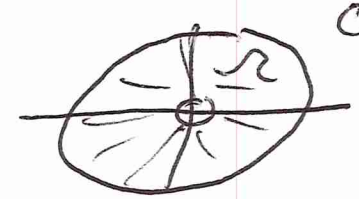
<u>Aside</u> If domain does not circle the origin then there are other sol'ns e.g.

$$u(x,y) = \theta = \tan^{-1}\left(\frac{x}{y}\right)$$

$$u(r,\theta) = r^{1/2}\cos\left(\tfrac{1}{2}\theta\right)$$



does not



circles origin

$n \geq 3$ ~~we~~ Consider spherical coord (r, θ, ϕ)

$$\Delta u = \frac{1}{r^{n-1}} \frac{\partial}{\partial r}\left(r^{n-1} \frac{\partial u}{\partial r}\right) + \frac{1}{r^2} \Delta_n u = 0$$

Look for sol'ns of the form

$$u(r, \theta, \phi) = R(r)\, Y(\theta, \phi)$$

As before

$$(r^2 R')' - \mu R = 0$$

Look for indep. ~~variables~~ sol'ns

r^α

$$\Delta_3 Y + \mu Y = 0$$

eq'n on unit sphere

Y needs to be periodic in θ + ϕ + continuous as approach the North + South poles i.e. at $\phi = 0, \pi$. This is always an issue numerically.

S 7.3

What change of indep. variables ~~gi~~ still give us harmonic f'ns.

<u>Examples</u> 1) Translation $x' = x + x_0$, $y' = y + y_0$ Preserve harmonicity

Linear transformats

$\underline{x} = \underline{\underline{A}}\, \underline{x}'$

preserves if

$\underline{\underline{A}} = \lambda \underline{\underline{I}}$ or

$\underline{\underline{A}} = \lambda \underline{\underline{B}}$ where

$\underline{\underline{B}}^T \underline{\underline{B}} = \underline{\underline{I}}$

↑ orthogonal

2) Rotations

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Preserve harmonicity

3) Reflections $x' = x$, $y' = -y$

4) Stretching $x' = \lambda x$ $y' = \lambda y$

only if same

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

How to show a transformation preserves harmonicity?

Change of variables:

$x' = x + x_0 \quad y' = y + y_0$

$x = x' - x_0 \quad y = y' - y_0$

$$\frac{\partial u}{\partial x'} = \frac{\partial u}{\partial x}\underbrace{\frac{dx}{dx'}}_{1} + \frac{\partial u}{\partial y}\underbrace{\frac{dy}{dy'}}_{1}$$ keep going.

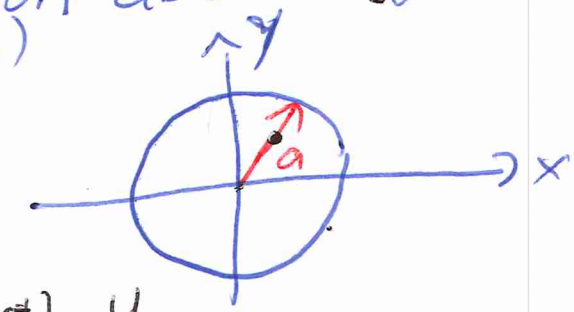
$u(r,\theta) = \ln(x^2+y^2)^{1/2}$ is harmonic

$\ln((x-2)^2+(y+5)^2)^{1/2}$ is also harmonic.

5) Preserves harmonicity: ~~reflection~~ (inversion) about sphere

$(r,\theta) \to (r^*, \theta^*)$

$\theta^* = \theta, \quad r r^* = a^2$



ex) $a = 2$ $\quad r = 1 \quad r^* = 4 \quad (1)(r^*) = 4$

$r = 2 \quad r^* = 2$

$r = 0 \quad r^* = \infty$

§7.4 Please read. $\Delta u = 0$

Review of Dirichlet BC, Neumann BC, + mixed BC.

$\downarrow$ Prescribe u

$\downarrow$ Prescribe $\frac{\partial u}{\partial n} = \underline{\nabla} u \cdot \underline{n}$ "flux"

$\downarrow$ $\alpha u + \beta \frac{\partial u}{\partial n} = g(\underline{x})$

$\underline{x} \in \partial\Omega$.